Pace Analytical Services, LLC
4171 40th St. SE
Grand Rapids, MI 49512
(616)975-4500

May 13, 2026

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Total Cyanide
Pace Project No.: 50430430

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on May 05, 2026. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Indianapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures

cc:

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CERTIFICATIONS

Project: Total Cyanide
Pace Project No.: 50430430

Pace Analytical Services Indianapolis

7726 Moller Road, Indianapolis, IN 46268	Illinois Accreditation #: 200074
Indiana Drinking Water Laboratory #: C-49-06	Kansas/TNI Certification #: E-10177
Kentucky UST Agency Interest #: 80226	Kentucky WW Laboratory ID #: 98019
Louisiana Certification #: 04076	Michigan Drinking Water Laboratory #9050
Oklahoma Laboratory #: 9204	Texas Certification #: T104704355
Wisconsin Laboratory #: 999788130	USDA Compliance Agreement #: IN-SL-22-001

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SAMPLE SUMMARY

Project: Total Cyanide
Pace Project No.: 50430430

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50430430001	**Influent**	Water	05/04/26 11:22	05/05/26 10:15
50430430002	**Effluent**	Water	05/04/26 11:16	05/05/26 10:15

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SAMPLE ANALYTE COUNT

Project: Total Cyanide
Pace Project No.: 50430430

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50430430001	**Influent**	SM 4500-CN-E	JTR	1	PASI-I
50430430002	**Effluent**	SM 4500-CN-E	JTR	1	PASI-I

PASI-I = Pace Analytical Services - Indianapolis

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SUMMARY OF DETECTION

Project: Total Cyanide
Pace Project No.: 50430430

Lab Sample ID / Method	Client Sample ID / Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50430430001	**Influent**					
SM 4500-CN-E	Cyanide	0.0024J	mg/L	0.0050	05/12/26 16:33	
50430430002	**Effluent**					
SM 4500-CN-E	Cyanide	0.0016J	mg/L	0.0050	05/12/26 16:34	

Date: 05/13/2026 12:51 PM

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ANALYTICAL RESULTS

Project: Total Cyanide
Pace Project No.: 50430430

Sample: Influent **Lab ID: 50430430001** Collected: 05/04/26 11:22 Received: 05/05/26 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
4500CNE Cyanide	Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis								
Cyanide	**0.0024J**	mg/L	0.0050	0.00070	1	05/11/26 10:07	05/12/26 16:33	57-12-5	

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REPORT OF LABORATORY ANALYSIS

Date: 05/13/2026 12:51 PM

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ANALYTICAL RESULTS

Project: Total Cyanide
Pace Project No.: 50430430

Sample: Effluent **Lab ID: 50430430002** Collected: 05/04/26 11:16 Received: 05/05/26 10:15 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
4500CNE Cyanide	Analytical Method: SM 4500-CN-E Preparation Method: SM 4500-CN-C Pace Analytical Services - Indianapolis								
Cyanide	**0.0016J**	mg/L	0.0050	0.00070	1	05/11/26 10:07	05/12/26 16:34	57-12-5	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Total Cyanide
Pace Project No.: 50430430

QC Batch: 897283
QC Batch Method: SM 4500-CN-C
Analysis Method: SM 4500-CN-E
Analysis Description: 4500CNE Cyanide
Laboratory: Pace Analytical Services - Indianapolis

Associated Lab Samples: 50430430001, 50430430002

METHOD BLANK: 4114387 Matrix: Water

Associated Lab Samples: 50430430001, 50430430002

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
Cyanide	mg/L	<0.00070	0.0050	0.00070	05/12/26 16:18	

LABORATORY CONTROL SAMPLE: 4114388

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/L	0.1	0.090	90	90-110	

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 4114389 4114390

Parameter	Units	50430132003 Result	MS Spike Conc.	MSD Spike Conc.	MS Result	MSD Result	MS % Rec	MSD % Rec	% Rec Limits	RPD	Max RPD	Qualifiers
Cyanide	mg/L	<0.0050	0.1	0.1	0.092	0.085	92	84	90-110	9	20	M0

MATRIX SPIKE SAMPLE: 4114391

Parameter	Units	50430622002 Result	Spike Conc.	MS Result	MS % Rec	% Rec Limits	Qualifiers
Cyanide	mg/L	0.0049J	0.1	0.094	89	90-110	M0

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALIFIERS

Project: Total Cyanide
Pace Project No.: 50430430

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.
ND - Not Detected at or above adjusted reporting limit.
TNTC - Too Numerous To Count
J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.
MDL - Adjusted Method Detection Limit.
PQL - Practical Quantitation Limit.
RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.
S - Surrogate
1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.
Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.
LCS(D) - Laboratory Control Sample (Duplicate)
MS(D) - Matrix Spike (Duplicate)
DUP - Sample Duplicate
RPD - Relative Percent Difference
NC - Not Calculable.
SG - Silica Gel - Clean-Up
U - Indicates the compound was analyzed for, but not detected.
N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.
Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.
Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.
TNI - The NELAC Institute.

ANALYTE QUALIFIERS

M0 Matrix spike recovery and/or matrix spike duplicate recovery was outside laboratory control limits.

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Total Cyanide
Pace Project No.: 50430430

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50430430001	**Influent**	SM 4500-CN-C	897283	SM 4500-CN-E	897663
50430430002	**Effluent**	SM 4500-CN-C	897283	SM 4500-CN-E	897663

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COOLER #63

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

CHAIN-OF-CUSTODY Analytical Request Document

Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

LAB USE ONLY - Affix Workorder/Login Label Here

WO# : 50430430

50430430

Company Name:	City of Cadillac	Contact/Report To:	Cindy Tomaszewski
Street Address:	1121 Plett Rd, Cadillac, MI 49601	Phone #:	231-775-2368
		E-Mail:	ctomaszewski@cadillac-mi.net
		Cc E-Mail:	
Customer Project #:		Invoice To:	Cindy Tomaszewski
Project Name:	Total Cyanide	Invoice E-Mail:	ctomaszewski@cadillac-mi.net
Site Collection Info/Facility ID (as applicable):		Purchase Order # (if applicable):	
		Quote #:	
Time Zone Collected: [ ] AK [ ] PT [ ] MT [ ] CT [X] ET		County / State origin of sample(s):	Michigan

Data Deliverables: [ ] Level II [ ] Level III [ ] Level IV [ ] EQUIS [ ] Other

Regulatory Program (DW, RCRA, etc.) as applicable: NPDES — Reportable [ ] Yes [ ] No

Rush (Pre-approval required): [ ] Same Day [ ] 1 Day [ ] 2 Day [ ] 3 Day [ ] Other ______

DW PWSID # or WW Permit # as applicable:

Date Results Requested:

Field Filtered (if applicable): [ ] Yes [X] No

Analysis:

* Matrix Codes (Insert in Matrix box below): Drinking Water (DW), Ground Water (GW), Waste Water (WW), Product (P), Soil/Solid (SS), Oil (OL), Wipe (WP), Tissue (TS), Bioassay (B), Vapor (V), Surface Water (SW), Sediment (SED), Sludge (SL), Caulk (CK), Leachate (LL), Biosolid (BS), Other (OT)

Specify Container Size **

Identify Container Preservative Type***

Analysis Requested

**Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) EnCore, (8) TerraCore, (9) 90mL, (10) Other

*** Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Lab Use Only
Proj. Mgr: **Brian Hall**
AcctNum / Client ID:
Table #:
Profile / Template: **7675**
Prelog / Bottle Ord. ID: **EZ 3084076**

Preservation non-conformance identified for sample.

Customer Sample ID	Matrix *	Comp / Grab	Composite Start Date	Composite Start Time	Collected or Composite End Date	Collected or Composite End Time	# Cont.	Res. Chlorine Results	Res. Chlorine Units	4500CNE Cyanide	Sample Comment
Influent	WW	G			5-4-26	1122	1	–	–	X	-001
Effluent	WW	G			5-4-26	1116	1	–	–	X	-002

Additional Instructions from Pace®:

Collected By: (Printed Name) Taysha Laczko

Signature: T. Laczko

Customer Remarks / Special Conditions / Possible Hazards:

# Coolers:	Thermometer ID:	Correction Factor (°C):	Obs. Temp. (°C)	Corrected Temp. (°C)	On Ice:

Relinquished by/Company: (Signature)	Date/Time:	Received by/Company: (Signature)	Date/Time:
T. Laczko City of Cadillac	5-4-26 1130	Ship VIA UPS Ground	
		Pace	5-5-26 1015

Tracking Number:

Delivered by: [ ] In-Person [ ] Courier [ ] FedEX [ ] UPS [ ] Other

Page: 1 of 1

Submitting a sample via this chain of custody constitutes acknowledgment and acceptance of the Pace® Terms and Conditions found at https://www.pacelabs.com/resource-library/resource/pace-terms-and-conditions/

ENV-FRM-CORQ-0019_v02_110123 ©

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Pace

Sample Conditions Upon Receipt Form (SCUR)

WO#: 50430430
PM: BJH **Due Date: 05/19/26**
CLIENT: GR-Cadillac

Date/Time: 5/5/26	**Evaluated By:** DN			
Client: City of Cadillac	**PM:** BJH			
Lab Notified of Rush or Short Holds:	YES NO ~~NA~~			
Project Received Via: FedEx ~~UPS~~ Client ~~Pace Courier~~ Other: ______ DN 5/5/26				**Comments:**
Custody Seal Present and Intact:	YES	NO	~~NA~~	
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	~~NA~~	
Short Hold Present (≤ 48 Hours):	YES	~~NO~~		
Sample Received in Hold:	~~YES~~	NO		
Custody Signature Present:	~~YES~~	NO		
Collector Signature Present:	~~YES~~	NO		
Sample Collected Today and On Ice:	YES	NO	~~N/A~~	
IR Gun #: ~~350~~ 351 **Therm #:** 282 283	**Temp. should be 0°C - 6°C (Initial/Corrected)**			
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: 4.2/4.0 °C			
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: ______ °C			
Temp Blank Received:	YES	~~NO~~		
Sample Label Matches COC (ID/Date/Time):	~~YES~~	NO		
Container Intact:	~~YES~~	NO		
Correct Container:	~~YES~~	NO		
Sufficient Volume:	~~YES~~	NO		
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with pink sticker on the lid if unacceptable. **pH Strip Lot #:** HC574455 *Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl*	~~YES~~	NO	N/A	
Residual Chlorine Absent: **Cl_2 Strip Lot #:** HC460768 *Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide*	~~YES~~	NO	~~N/A~~	DN 5/5/26
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	~~N/A~~	
Trip Blank Received: HCl MeOH Other: ______	YES	~~NO~~	ON HOLD	
Comments:	3. Cooler Temp. Upon Receipt: ______ °C			
	4. Cooler Temp. Upon Receipt: ______ °C			
	Non-Conformance Form Required: YES ~~NO~~			

F-GR-C-007-rev.09, 11Aug2025

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